

L-5: Standard Nowcasting Methods

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Introduction

- Policy-makers need to assess the current state of the economy in real-time, when only incomplete information is available.
- In real-time, unbalanced datasets arise mainly due to two reasons:
  - the different sampling frequencies with which the variables are available
  - the "ragged-edged" problem: publication delays of variables that cause missing values of some of the variables at the end of the sample
- Example: Gross Domestic Product (GDP) is released quarterly and with considerable publication lag, while a range of leading and coincident variables is available in a timelier fashion, and at a monthly or even higher frequency.

Introduction

Two common solutions to handle the mixed frequency problem:

1. Temporally aggregate all data to the same (low) frequency by computing sums or averages or by taking the last high frequency observations in each low frequency period (point-in-time or stock sampling).

   Example: sum m-o-m growth rates of industrial production (IP) over 3 months to get q-o-q IP growth rate that can be compared with q-o-q GDP growth.

But temporal aggregation creates various problems (Marcellino (1999)):

- Loss of efficiency (not all the observations are used, increasing the uncertainty around parameter estimates)
- Loss of specific econometric features such as exogeneity (due to changes in the dynamics of the aggregated system)
- Loss of parameters (due to aggregation of original parameters)

Introduction

2. Temporally interpolate all data to the same (high) frequency. Various methods are available (e.g., Chow-Lin, Denton, splines, etc (see Angelini, Henry and Marcellino (2006))

   Example: quarterly GDP growth can be split into 3 monthly m-o-m growth rates, which in each quarter add up to the original quarterly growth rate

But interpolation also creates various problems:

- Generation of measurement errors
- Possible collinearity problems when the same indicators are used both for interpolation and then in the high frequency model (e.g., monthly IP used to interpolate quarterly GDP, and then monthly IP included in a VAR with the interpolated GDP, which now has a monthly frequency).
- Lack of theoretical correspondence between disaggregated and aggregated models.

Introduction

The theoretically optimal method to handle mixed frequencies (and data irregularities such as missing values):

State-space approach:

- It casts a high frequency model in state space form
- It treats low frequency variables as high frequency variables with missing observations (e.g., in a monthly model, quarterly GDP growth rates are only available in March, June, etc., and missing in January, February, April, May, etc.)
- It uses the Kalman filter to get optimal estimates of missing observations, and optimal forecasts/nowcasts of all variables
- Examples: mixed-frequency VARs and factor models (e.g., Mariano & Murasawa, (2003, 2010), Giannone et al (2008))

Some issues:

- Requires one to know the high frequency model
- Complex when the model is not linear
- Computationally demanding when frequency mismatches are large (e.g., daily financial data and quarterly macro indicators)

Introduction

Simpler methods to handle mixed frequency data:

1. Bridge models (e.g. Baffigi et al, 2004)

- Linear regressions that link ("bridge") high frequency variables to low frequency ones (e.g., monthly IP and quarterly GDP)
- The inclusion of specific RHS variables/indicators is not based on causal relations, but on the statistical fact that they contain timely updated information
  - First, monthly indicators are forecast over the remainder of the quarter, usually based on univariate time series models, then aggregated to obtain their corresponding quarterly values
  - Second, the aggregated (low frequency) values are used as regressors in the bridge equation which allows to obtain forecasts of the low frequency target variable

Introduction

2. MIDAS approach (e.g., Ghysels et al (2004) and Clements and Galvao (2008))
   - Generalization of "Almon lag" polynomial approach (non-linear) described below
   - Tightly parameterized, reduced-form regression that involves processes sampled at different frequencies
   - Can handle large frequency mismatch by reducing the number of parameters to be estimated
   - Direct forecasting tool
3. Unrestricted MIDAS (U-MIDAS) approach (e.g., Foroni et al (2015))
   - Generalization of bridge equation and MIDAS regressions
   - Linear model to which we can apply OLS estimation
   - Cannot handle large frequency mismatch
   - Can be applied also to factor models

Bridge Equations

Let y be the low frequency (LF) target variable (e.g., quarterly GDP growth), and x_i the high frequency (HF) indicators, $i = 1, \ldots, j$ (e.g., monthly industrial production, survey data, etc.), which are aggregated to LF according to their stock/flow nature.

$t_q = 1, \ldots, T$ indicates time in quarters, t_m in months.

- The bridge model to be estimated is:

$$y_{t_q} = \alpha + \sum_{i=1}^{j} \beta_i(L) x_{it_q} + u_{t_q}$$

$\beta_i(L)$ are polynomials in the lag operator L

$\beta_i(L) = \beta_{0i} + \beta_{1i}L + \cdots + \beta_{p_i i}L^{p_i}$, one for each HF indicator, where p_i is the number of lags

u_{t_q} is an i.i.d. error term

Bridge Equations

- The target frequency t_q is quarterly, but the model can be re-estimated each month. A nowcast is constructed as :

$$\hat{y}_{T+1} = \hat{\alpha} + \sum_{i=1}^{j} \hat{\beta}_i(L) x_{iT+1}$$

$\hat{\alpha}$ and $\hat{\beta}_{si}$, s= $0, \ldots, p$ are OLS estimators.

- When not all months of x_i in quarter $T+1$ are observable, we need to forecast the missing monthly observations prior to aggregation into x_{iT+1} :
  - Use AR/ARMA models (with lag length chosen by an information criterion).

    Example: at the end of month 2 of quarter $T+1$ we could have observations up to month 1 for x_1 (IP) and up to month 2 for x_2 (survey data).

Bridge Model: Example

- Suppose we are interested in March GDP (which is published only on a quarterly basis and at the end of the quarter) and that we base our computation only on IP. Suppose we are currently in January and we observe January IP (which is published on a monthly basis and at the end of the month).

- We make a one month ahead forecast of IP to get the February IP forecast, then make a two month ahead forecast of IP to get a March IP forecast, and we aggregate the January observation with the 2 IP forecasts to get a quarterly forecast value of IP.

- We then run a regression of GDP on IP up until December to get the coefficient estimate on IP (where IP is viewed as a quarterly variable through aggregation but with no need for forecast of IP as IP up to December of the previous year is published). Then add as last value of the IP series the aggregated IP variable just constructed (including the forecast of IP for February and March), and produce a forecast of GDP using the estimated coefficient on IP.

NOTE: The forecast of GDP will change (i.e. will be revised) on a monthly basis as the IP series is updated to include the new released IP data.

Bridge Equations

- The lag length of $\beta_i(L)$, p_i, is often set to zero, so that $\beta_i(L) = \beta_{0i}$ and the model is static:

$$y_{t_q} = \alpha + \sum_{i=1}^{j} \beta_{0i} x_{it_q} + u_{t_q}$$

This facilitates economic interpretation of the results.

- p_i can also be chosen according to an information criterion.
- As the Bridge equation is linear, OLS estimation is optimal, and the optimal nowcast/forecast (assuming a quadratic loss function) is obtained by replacing the unknown parameters with the OLS estimates.
- Parameter time-variation or specific forms of non-linearity can be introduced → U-MIDAS models.

MIDAS Models: Motivation

Example: Suppose we have monthly data on the money supply, and we would like to use it to forecast quarterly GDP.

- Use the January, April, July, October money supply data to create a first quarterly variable of money supply.
- Then use February, May, August and November to create a second quarterly variable of money supply.
- Finally, use March, June, September and December.

Then run a regression of quarterly GDP on the three quarterly variables of money supply.

- We could also run a VAR using the quarterly variables created, but dimensionality issues arise very quickly.

NOTE:

This is different from Bridge where, for example, we use aggregation of monthly data within a quarter to create a quarterly value.

MIDAS Models

- MIDAS models (Ghysels et al (2004)) are tightly parameterized reduced form regressions with variables sampled at different frequencies.
  - The response of the dependent variable (e.g., GDP) to the HF explanatory variable (e.g., interest rate, money supply etc.) is modelled using distributed lag polynomials with coefficients depending on a small number of parameters, to prevent parameter proliferation when frequency mismatch is large (e.g., daily/quarterly)
- The basic MIDAS model for a single explanatory variable, and h_q-step-ahead forecasting, with $h_q = h_m/m$, is:

$$y_{t_q+mh_q} = y_{t_m+h_m} = \beta_0 + \beta_1 b(L_m;\theta) x_{t_m+w}^{(m)} + \epsilon_{t_m+h_m}$$

$$b(L_m;\theta) = \sum_{k=0}^{K-1} c(k;\theta) L_m^k \qquad L_m^s x_{t_m}^{(m)} = x_{t_m-s}^{(m)}$$

K is the aggregation frequency (e.g., K=3 in the monthly to quarterly case)

$x_{t_m+w}^{(m)}$ is skip-sampled from the HF indicator x_{t_m}

Interpretation

- h_m represents the number of months of the forecast period (i.e., the length) . Hence $h_m = mh_q$, where m is the number of months in a quarter (i.e. 3).
- $x^{(m)}$ represents the HF frequency variable – the (m) indicates the sampling frequency relative to the base frequency (in this case quarterly).
  - if the base frequency is quarterly, then (m) = (3), that is – there are three x's associated with a quarterly observation.
- $x_{t_m+w}^{(m)}$: if w = 0 and the quarterly data is positioned at the end of the quarter, then we are referring to the quarterly series derived from the monthly indicator that has the March, June, September, and December observations in it. In general w (an integer) indicates the number of months that exist between when y_{t_q} (quarterly dependent variable) was published and the actual date (i.e. position in the quarter) of the variable $x^{(m)}$
  - For example, suppose quarterly GDP, as published, is for the last month of the quarter. And suppose that the high frequency variable is for January, April, July, and October. Then the offset is w = -2 months (monthly data is published 2 months before GDP).
  - If w = 0, then the monthly data would be for March, June, September and December.
  - In short, w represents the number of months that the HF indicator is available before the publication of quarterly GDP for a particular quarter.

Interpretation

Assume quarterly GDP data are positioned at the end of each quarter (March, June, September, and December).

- The quarterly variable created from the monthly HF variable that contains the March, June, September and December values has a (monthly) lag of 0 relative to quarterly GDP.
    - Similarly, the quarterly variable created from the monthly HF variable that contains February, May, August, and November values has a (monthly) lag of 1 relative to quarterly GDP.
    - The quarterly variable created from the monthly HF variable that contains January, April, July, and October values has a (monthly) lag of 2 relative to quarterly GDP.

This is the role of the b( ) function: It creates variables from the high frequency variable via appropriate lag operations on the HF variable, together with skip sampling (e.g., taking every 3rd observation in the case of monthly to quarterly).

Interpretation

- $b(L_m;\theta)$ is a polynomial in the lag operator L for the high frequency variable (e.g., m here refers to the case where the HF is monthly).

- "skip-sampling" refers to selecting a specific sequence of months to create the RHS x variables.
    - For monthly to quarterly, assuming w = -2, we would take the following observations in the monthly data: 1, 4, 7, 10,... (thereby skipping 2 and 3, 5 and 6, etc.).
        - If w = -1, we would select 2, 5, 8, 11, ...
        - If w = 0, we would use 3, 6, 9, 12...

MIDAS Models

$$y_{t_q+mh_q} = y_{t_m+h_m} = \beta_0 + \beta_1 b(L_m;\theta) x^{(m)}_{t_m+w} + \epsilon_{t_m+h_m}$$

$$b(L_m;\theta) = \sum_{k=0}^{K-1} c(k;\theta) L_m^k \qquad L_m^s x^{(m)}_{t_m} = x^{(m)}_{t_m-s}$$

- The coefficients $c(k;\theta)$ in MIDAS models typically depend on many unknowns, leading to the curse of dimensionality (especially with mixed frequency VARs). One of the most used parameterizations is the "Exponential Almon Lag":

$$c(k;\theta) = \frac{\exp(\theta_1 k + \cdots + \theta_Q k^Q)}{\sum_{k=0}^{K-1} \exp(\theta_1 k + \cdots + \theta_Q k^Q)}$$

- This function is quite flexible and can take various shapes with only a few parameters (often *Q=2*), helping to avoid the high dimensionality issue.
- This function can include decreasing, increasing or hump-shaped patterns.

Example of Exponential Almon

$$c(k;\theta) = \frac{\exp(\theta_1 k + \cdots + \theta_Q k^Q)}{\sum_{k=0}^{K-1} \exp(\theta_1 k + \cdots + \theta_Q k^Q)}$$

- The denominator is to ensure the weight sum to 1.
- Suppose you want to generate 10 lags from 2 parameters. How do you do that?

$Lag\ 0\ (k=0)\colon\ c(0;\theta) = exp(0) = 1$

$Lag\ 1\ (k=1)\colon\ c(1;\theta) = exp(\theta_1 + \theta_2)$

$Lag\ 2\ (k=2)\colon\ c(2;\theta) = exp(2\theta_1 + 4\theta_2)$

$Lag\ 3\ (k=3)\colon\ c(3;\theta) = ex\,p(3\theta_1 + 9\theta_2)$ ...

$Lag\ 10\ (k=10)\colon\ c(10;\theta) = exp(10\theta_1 + 100\theta_2)$

$Denom = 1 + ex\,p(\theta_1 + \theta_2) + ex\,p(2\theta_1 + 4\theta_2) + .. + exp(10\theta_1 + 100\theta_2)$

MIDAS Models

- Another possible parameterization, also with only two parameters, is the so-called "Beta Lag", because it is based on the Beta function:

$$c(k;\theta_1,\theta_2) = \frac{f(\frac{k}{K},\theta_1;\theta_2)}{\sum_{k=1}^{K} f(\frac{k}{K},\theta_1;\theta_2)}$$

$$f(x,a;b) = \frac{x^{a-1}(1-x)^{b-1}\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$$

$$\Gamma(a) = \int_0^{\infty} e^{-x}x^{a-1}dx$$

MIDAS Models

Ghysels, Rubia and Valkanov (2009) propose 3 other different parameterizations:

- A linear scheme with $c(k;\theta) = \frac{1}{K}$ where there are no parameters to estimate in the lagged weight function
- An hyperbolic scheme, with $c(k;\theta) = \frac{g(\frac{k}{K},\theta)}{\sum_{k=0}^{K-1} g(\frac{k}{K},\theta)}$, $g(k,\theta) = \frac{\Gamma(k+\theta)}{\Gamma(k+1)\Gamma(\theta)}$

  where the gamma function has only one parameter to estimate (but it is not as flexible as the Beta specification)
- A geometric scheme with $c(k;\theta) = \frac{\theta^k}{\sum_{k=0}^{\infty}\theta^k}$, $|\theta| \leq 1$

  and $c(k;\theta)$ are normalized so that they sum to one

MIDAS Models

- The parameterizations for $c(k;\theta)$ are all quite flexible: for different values of the parameters θ, they can take various shapes:
  - weights attached to the different lags can decline slowly or fast, or even have a hump shape.

  → estimating the parameters from the data automatically determines the shape of the weights and the number of lags to be included in the regression.
- The MIDAS model can be estimated using nonlinear least squares (NLS) in a regression of y_t onto $x_{t-h}^{(m)}$
- The nowcast/forecast is

$$\hat{y}_{T_m^y+h_m|T_m^x} = \hat{\beta}_0 + \hat{\beta}_1 b(L_m;\hat{\theta})x_{T_m^x}^{(m)}$$

Note: MIDAS is h_m-dependent and needs to be re-estimated for each forecast horizon ➔ for a given choice of h_m , we will obtain different estimates of the model parameters, since we are projecting on a different information set (as usual in direct forecasting)

MIDAS AR Models

Since autoregressive (AR) models often provide competitive forecasts, the introduction of an AR term in the MIDAS model is desirable, but it is not straightforward.

- Consider adding a low frequency lag of y_{t_m}, y_{t_m-3} , to the basic model with K =3 (x is monthly and y is quarterly):

$$y_{t_m} = \beta_0 + \lambda y_{t_m-3} + \beta_1 b(L_m;\theta)x_{t_m+w}^{(3)} + \epsilon_{t_m}$$

The model can be written as:

$$y_{t_m} = \beta_0(1-\lambda)^{-1} + \beta_1 (1-\lambda L_m^3)^{-1} b(L_m;\theta)x_{t_m+w}^{(3)} + (1-\lambda L_m^3)^{-1}\epsilon_{t_m}$$

The polynomial on $x_{t_m+w}^{(3)}$ is the product of a polynomial in L_m and a polynomial in L_m^3. Clements and Galvao (2009) note that this generates a seasonal response of y to $x^{(3)}$, irrespective of whether $x^{(3)}$ displays a seasonal pattern.

MIDAS AR Models

- Clements and Galvao (2009) suggest the introduction of the AR dynamics as a common factor:

$$y_{t_m} = \beta_0 + \lambda y_{t_m-3} + \beta_1 b(L_m;\theta)(1-\lambda L_m^3) x^{(3)}_{t_m+w-3} + \epsilon_{t_m}$$

so that the response of y to $x^{(3)}$ remains non-seasonal.

- The analogous multi-step model is written as:

$$y_{t_m} = \beta_0 + \lambda y_{t_m-h_m} + \beta_1 b(L_m;\theta)\left(1-\lambda L_m^{h_m}\right) x^{(h_m)}_{t_m+w-h_m} + \epsilon_{t_m}$$

MIDAS AR Models

- To estimate these models: the common procedure is to estimate the standard MIDAS (the basic model), take the associated residuals $\hat{\epsilon}_{t_m}$ and estimate an initial value for λ: $\hat{\lambda}_0 = (\sum \hat{\epsilon}^2{}_{t_m+w-h_m})^{-1} \sum \hat{\epsilon}_{t_m} \hat{\epsilon}_{t_m+w-h_m}$
- Then construct:

$$y^*_{t_m} = y_{t_m} - \hat{\lambda}_0 y_{t_m-h_m}$$

$$x^{*(3)}_{t_m+w-h_m} = x^{(3)}_{t_m+w-h_m} - \hat{\lambda}_0\, x^{(3)}_{t_m-(h_m-w)-h_m}$$

- The estimate $\hat{\theta}_1$ is obtained by applying nonlinear least squares to

$$y^*_{t_m} = \beta_0 + \beta_1 b(L_m;\theta) x^{*(3)}_{t_m+w-h_m} + \epsilon_{t_m}$$

A new value of λ, $\hat{\lambda}_1$ is obtained from the residuals of this regression. Then another step is run, using $\hat{\lambda}_1$ and $\hat{\theta}_1$ as the initial values. In this way, the procedure gets estimates of $\hat{\lambda}$ and $\hat{\theta}$ that minimize the sum of squared residuals.

The U-MIDAS approach

- Foroni, Marcellino and Schumacher (2015) study the performance of a variant of MIDAS which does not resort to functional distributed lag polynomials:
  - They discuss how unrestricted MIDAS (U-MIDAS) regressions can be derived in a general linear dynamic framework, and under which conditions the parameters of the underlying high-frequency model can be identified.
- The U-MIDAS model can be written:

$$c(L_m^m)y_{t_m} = \delta_1(L_m)x_{1t_m} + \ldots + \delta_N(L_m)x_{Nt_m} + \epsilon_{t_m}$$

$$c(L_m^m) = (1 - c_1 L_m^m - \cdots - c_s L_m^{ms})$$

$$\delta_j(L_m) = \delta_{j,0} + \delta_{j,1}L_m + \cdots + \delta_{j,v}L_m^v, \qquad j = 1, \ldots, N$$

NOTE: The U-MIDAS model is estimated in LF (as MIDAS and Bridge), uses HF regressors (as MIDAS, not as Bridge), and can be re-estimated each month within the quarter. As the U-MIDAS model is linear, it can be estimated by OLS.

The U-MIDAS approach

- A simple approach to forecasting is to first estimate the model (e.g. with OLS) and then construct the forecast as:

$$\tilde{y}_{T_m^x+m|T_m^x} = \tilde{c}(L_m^m)y_{T_m^x} + \widetilde{\delta_1}(L_m)x_{1T_m^x} + \cdots + \widetilde{\delta_N}(L_m)x_{NT_m^x}$$

The polynomials:

$$\tilde{c}(.) = \widetilde{c_1}\, L_m^m + \cdots + \widetilde{c_s}\, L_m^{ms}$$

$$\widetilde{\delta_i}(L_m)$$

are obtained by regressing y_{t_m} on information dated $mt_m - m$ or earlier, for $t_m = 1,2, \ldots, T_m^x$

The U-MIDAS approach

- In general, the direct approach can be extended to construct h_m-step ahead forecasts given information in T_m^x :

$$\bar{y}_{T_m^x+h_m|T_m^x} = \bar{c}(L_m^m)y_{T_m^x} + \overline{\delta_1}\,(L_m)x_{1T_m^x} + \cdots + \overline{\delta_N}(L_m)x_{NT_m^x}$$

The polynomials $\bar{c}(.)$ and $\overline{\delta_i}(L_m)$ are obtained by regressing $\bar{y}_{t_m}$ on information dated $mt_m - h_m$ or earlier, for $t_m = 1,2,\ldots,T_m^x$

NOTE: In the case of U-MIDAS, and as for Bridge, an autoregressive term can be included easily without any restrictions on its coefficient.

Simplest Form of U-MIDAS

- In its simplest form, the U-MIDAS model is a regression of y (real GDP at the quarterly frequency) on the 3 monthly HF skip-sampled variables:

$$y_{t_m} = \beta_0 + \beta_1 x_{1t_m}^{(m)} + \beta_2 x_{1t_m-1}^{(m)} + \beta_3 x_{1t_m-2}^{(m)} + \epsilon_{t_m}$$

- There are no lagged dependent variables in the equation, and no lags (in terms of quarters) of the skip-sampled HF variable, $x_{1t_m}^{(m)}$.

- U-MIDAS representations can be used when the number of skip-sampled HF variables on the RHS is small. The unknown parameters can be estimated directly using OLS, and forecasts obtained from:

$$\hat{y}_{T_m^x+h_m|T_m^x} = \hat{\beta}_0 + \hat{\beta}_1 x_{1T_m^x}^{(m)} + \hat{\beta}_2\, x_{1T_m^x-1}^{(m)} + \hat{\beta}_3 x_{1T_m^x-2}^{(m)}$$

Mixed-Frequency Factor Models

- There may be situations where many HF variables could be used, but mixed frequency models introduced so far may not be able to manage the large number of parameters to estimate.

- Idea: To allow for the inclusion of factors into the mixed frequency framework.

Step 1. Extract factors from the set of HF variables.
  - This can be done using Principal Components Analysis (PCA) or 3 Pass Regression Filter (3PRF).

Step 2. Proceed as before with MIDAS (or U-MIDAS or Bridge).

We will review these methods in greater detail in the next lecture.

Extensions: Multiple explanatory variables

- To allow for the inclusion of several additional explanatory variables into the MIDAS framework, we use:

$$y_{t_m} = \beta_0 + \beta_1\, b(L_m;\theta_1) x^{(m)}_{1,\,t_m+w-h_m} + \beta_2 b(L_m;\theta_2) x^{(m)}_{2,\,t_m+w-h_m} + \epsilon_{t_m}$$

In this case, we consider x_1 and x_2 as two different explanatory variables.
  - The values of the theta parameters are assumed to take on independent values; they may have different lag lengths.
- The above specification may be extended to allow for the inclusion of more than two explanatory variables (or more lags), and for the presence of an autoregressive structure.
- The most general MIDAS linear regression model:

$$y_{t_m} = \beta_0 + \sum_{i=1}^{K}\sum_{j=1}^{L} b_{ij}(L_{m_i};\theta_i)\, x^{(m_i)}_{i,\,t_m+w-h_m} + \epsilon_{t_m}$$

Extensions: Multiple explanatory variables

- Within the more general framework, it is also possible to include explanatory variables at different frequencies, since each indicator is modelled with its own polynomial parameterization

  Example: quarterly GDP growth can be explained not only by monthly indicators but also by weekly financial variables

- In practice, adding explanatory variables substantially complicates estimation.

➢ An alternative, easier procedure, which often works well in practice is to work with single indicator MIDAS models and then pool the resulting forecasts.

Empirical Application: Nowcasting Euro Area GDP Growth

Foroni and Marcellino (2014):

- Compare different methods (small/large scale, pooling) to deal with mixed frequencies and ragged-edge structure

- Use a very large dataset to compare the forecasting performance for Euro Area GDP growth and its components with mixed frequency procedures

- Check temporal stability of the results: rolling/recursive estimation and subsample analysis

Empirical Application: Nowcasting Euro Area GDP growth

Data:

- Euro area quarterly and monthly series from Eurostat dataset of Principal European Economic Indicators (PEEI):
  - quarterly GDP and components: 1996Q1 - 2009Q2
  - 150 monthly indicators: Jan 1996 - Aug 2009
- Pseudo real time dataset (it replicates the differences in the availability of data)

Ragged-edge structure: Main publication lags

Main releases	Publishing lag	Frequency
HICP	1 month	monthly
PPI	2 months	monthly
Industrial production	2 months	monthly
Industrial new orders	2 months	monthly
Turnover index	2 months	monthly
Hours worked	2 months	monthly
Car registrations	2 months	monthly
Retail trade	2 months	monthly
Construction output	2 months	monthly
Business survey	current month	monthly
Business climate indicator	current month	monthly
Consumer survey	current month	monthly
Money supply	1 month	monthly
Exchange rates (average)	1 month	monthly
Interest rates (average)	1 month	monthly
Stock exhange indexes (average)	1 month	monthly
Unemployment	2 months	monthly
GDP: disaggregation of sectorial value added	1 quarter	quarterly
GDP: disaggregation from expenditure side	1 quarter	quarterly

Source: Foroni and Marcellino (2014)

Empirical Application: Nowcasting Euro Area GDP growth

- Recursively estimate and then now- and forecast Euro area GDP growth rate, with the first evaluation quarter fixed at 2003Q1 and the last at 2009Q1 (25 recursive evaluation samples).
- In each recursion, predict the current GDP growth and the values of one and two quarters ahead.
- For every single quarter in the evaluation sample, there are 3 nowcasts and 6 forecasts.

  Example: for 2005Q3

  - nowcast computed in September 2005, in August 2005, in July 2005
  - one-quarter ahead forecasts computed in June, May and April 2005
  - two-quarters ahead forecasts computed in March, February and January 2005

Empirical Application: Nowcasting Euro Area GDP growth

Individual models are discussed for three nowcast horizons.

The paper uses MSE as a measure for comparing the performances of the different models. As a benchmark, they use a recursively estimated AR model for each component of GDP; the lag length is specified according to the BIC.

The results (shown next) display the average relative MSE performances of the different classes of mixed-frequency models, for each GDP component, at different horizons, against the AR benchmark.

Methodology: First estimate every individual model and compute the relative MSE with respect to the benchmark, i.e., calculate the MSE of every single indicator model relative to the MSE of the AR model. Then, average across all of the indicators within a model class (Bridge, MIDAS, AR-MIDAS and MF-VAR).

Empirical Application: Nowcasting Euro Area GDP growth

Average relative MSE performances of different classes of mixed-frequency models against AR benchmark: GDP components.

Component	Model	Horizon (h_m) 1	2	3
Production side				
Agriculture, hunting, forestry and fishing	bridge	**0.76**	**0.98**	**0.98**
	ar-midas	0.80	1.10	1.10
	mf-var	**0.76**	1.04	1.01
Total industry, excluding construction	bridge	1.06	0.90	**0.93**
	ar-midas	**0.92**	**0.89**	0.95
	mf-var	1.45	0.94	0.98
Construction	bridge	1.01	**1.07**	**1.05**
	ar-midas	**0.98**	1.10	1.09
	mf-var	1.02	1.95	1.07
Trade, hotels and restaurants, transport and communication services	bridge	1.10	1.10	1.11
	ar-midas	**1.05**	**0.92**	**0.94**
	mf-var	1.22	1.12	1.15
Financial services and business activities	bridge	0.84	0.83	0.87
	ar-midas	**0.83**	**0.77**	**0.81**
	mf-var	0.86	0.85	0.91
Other services	bridge	**1.00**	**1.01**	**1.01**
	ar-midas	1.09	1.20	1.15
	mf-var	1.15	1.14	1.11
Expenditure side				
Final consumption - households	bridge	**1.01**	0.96	0.93
	ar-midas	1.02	**0.85**	**0.85**
	mf-var	1.10	0.98	0.95
Final consumption - Government	bridge	**1.07**	**1.06**	**1.05**
	ar-midas	1.10	1.12	1.08
	mf-var	1.41	1.63	1.56
Gross fixed capital formation	bridge	1.08	1.04	1.05
	ar-midas	**0.93**	**0.92**	**0.95**
	mf-var	1.38	1.14	1.18
Imports	bridge	0.85	0.87	0.90
	ar-midas	**0.82**	**0.79**	**0.83**
	mf-var	1.03	0.91	0.96
Exports	bridge	0.94	0.88	0.92
	ar-midas	**0.91**	**0.86**	**0.91**
	mf-var	1.16	0.92	0.98

Notes: The entries in the table are obtained as follows: first, estimate every individual model recursively and compute the relative MSE with respect to the benchmark; then, take the average of the relative MSE within a model class across all of the indicators. The benchmark is the recursive estimate of an AR model with the lag length specified accordingly to the BIC. The evaluation sample is 2003Q1–2009Q1. The numbers in bold show the best relative MSE performance for each horizon and each component.

Empirical Application: Nowcasting Euro Area GDP growth

The evidence is generally in favor of the exploitation of monthly information when nowcasting the different GDP components

Example: GDP disaggregation from the production side

- The mixed-frequency approaches outperform the benchmark for those components for which several monthly indicators are available, as in the case of the industry sector and financial services.
- For agriculture, the lack of availability of monthly indicators is critical. The same is true for a variety of economic activities (public administration and defense, compulsory social security; education; health and social work; other community, social and personal service activities; private households with employed persons), for which it is not easy to find reliable and timely monthly indicators of the value they add.

General remark: the AR-MIDAS seems to work well for short horizons. The evidence in favor of MF-VAR is less strong, while the Bridge models perform well, beating the benchmark in many cases. The only method with some ability to beat the Bridge equations is AR-MIDAS.

Forecast pooling

- Implementing forecast combination allows overcoming mis-specification bias, parameter instability and measurement errors in the datasets which may be present in the individual forecasts (Timmermann, 2006).
- Combination schemes:
  - mean
  - median
  - weighted mean (weights inversely proportional to the MSE of the previous four-quarter performance of the model (Stock and Watson, 2001)

Forecast pooling: Performance of pooling within different classes of models vs AR benchmark

Relative MSE performance of model pooling within a given class of models against the AR benchmark (combination scheme: median).

Component	model	Horizon (h_m) 1	2	3	Component	Model	Horizon (h_m) 1	2	3
Production side					**Expenditure side**				
Agriculture, hunting, forestry and fishing	bridge	0.74*	**0.95**	**0.95**	Final consumption - households	bridge	0.97*	0.96*	0.97*
	ar-midas	0.75*	1.02	1.02		ar-midas	**0.92***	**0.71***	**0.79***
	mf-var	**0.69***	0.96	**0.95**		mf-var	0.95*	0.93*	0.96*
	all	0.73*	0.96	**0.95**		all	0.94*	0.85*	0.88*
Total industry, excluding construction	bridge	1.00*	0.96	0.96	Final consumption - Government	bridge	**1.00**	**1.00**	1.00*
	ar-midas	**0.90***	**0.89**	**0.95**		ar-midas	1.01	1.01	1.02
	mf-var	1.19	0.97*	1.01		mf-var	1.11	1.04*	1.03
	all	1.00	0.94*	0.96		all	1.01	**1.00**	**0.99**
Construction	bridge	0.92	**0.98**	**0.99**	Gross fixed capital formation	bridge	1.08	1.12	1.12
	ar-midas	0.93	1.03	1.04		ar-midas	**0.88***	**0.91**	**0.94**
	mf-var	**0.88**	0.99	1.00		mf-var	1.27	1.12	1.20
	all	0.92*	1.00	1.01		all	1.00	0.98	1.03
Trade, hotels and restaurants, transport and communication services	bridge	1.18*	1.22	1.23	Imports	bridge	0.87	0.93	0.94
	ar-midas	**1.01**	**0.91***	**0.94***		ar-midas	**0.81***	**0.76**	**0.79**
	mf-var	1.17	1.14	1.17		mf-var	0.96	0.89*	1.00
	all	1.09	1.02	1.05		all	0.86*	0.85*	0.89
Financial services and business activities	bridge	0.72*	0.74*	0.85	Exports	bridge	0.97	0.95	0.97
	ar-midas	0.73*	**0.66***	**0.77**		ar-midas	**0.90***	**0.83***	**0.90**
	mf-var	**0.71***	0.79*	0.92		mf-var	1.07	0.93*	1.01
	all	0.72*	0.74*	0.84		all	0.95*	0.90*	0.95
Other services	bridge	0.96*	**0.98**	**1.00**					
	ar-midas	**0.95**	0.99	1.03					
	mf-var	0.99	1.02	1.03*					
	all	**0.95**	0.99	1.01					

Notes: The entries in the table are obtained as follows: first, forecasts from single indicator models are computed, and the median of all of the forecasts within a single model is obtained; second, the MSE of the forecast combination is calculated and divided by the MSE of the benchmark. Only the last row of each component considers the median of all of the forecasts across methods. The estimation is conducted recursively. The benchmark is the recursive estimate of an AR model with the lag length specified according to the BIC. The evaluation sample is 2003Q1–2009Q1. The numbers in bold show the best relative MSE performance for each horizon and each indicator. Asterisks indicate the models which appear to be statistically superior to the benchmark at a confidence level of 10% according to the modified Diebold–Mariano test.

Forecast pooling: Performance of pooling within different classes of models vs AR benchmark

The table provides the relative MSE performance of model pooling within a given class of models against the benchmark, for each component of the GDP.

- First: forecasts from single indicator models are computed, and the median of all of the forecasts within a single approach is obtained.

- Second: the MSE of this forecast combination is calculated and divided by the MSE of the benchmark. Since we are comparing one forecast with the benchmark in this case, one can test the hypothesis of equal accuracy in forecasting performance.

Results:
- AR-MIDAS pooling performs fairly well: it outperforms the benchmark for most of the components at each nowcast horizon.
- Bridge models confirm their good performances across horizons.
- Pooling improves the performance for single indicators for the MF-VARs, but the gains with respect to the benchmark are smaller than for the other approaches.

Forecast pooling: Performance of pooling within different classes of models vs AR benchmark

The table contains the results of nowcast combinations of all models under consideration.

- In general, pooling all the models is not better than pooling within single classes of models.
- Despite the very short evaluation sample, which generally compromises a good performance of the test, there is evidence of the improvement in the forecast accuracy by using mixed-frequency data.
- The largest improvements upon the AR benchmark are for the components for which many indicators are available in the dataset.
    - For the other variables, there are still improvements (unlike in the single indicator case), but these are generally smaller.

Conclusion: Pooling mixed frequency models based on a large set of alternative indicators appears promising (this is in line with the forecast combination literature (Timmermann, 2006)).

Empirical Application: Nowcasting Euro Area GDP growth

Is the disaggregated information contained in the components of GDP useful for nowcasting total GDP growth?

In order to do this, aggregate the nowcasts obtained from the different components and compare them with the nowcasts obtained by considering only the total GDP aggregate.

Result: When using a simple constant weighting scheme, looking at disaggregate information can provide some improvements.

Conclusion

There are two common solutions to handle the mixed frequency problem:

1. Temporally aggregate all data to the same (low) frequency by computing sums or averages or by taking the last high frequency observations in each low frequency period.

2. Temporally interpolate all data to the same (high) frequency.

We noted that each solution has some advantages and disadvantages.

Conclusion

Simpler techniques available to work with mixed frequencies:

- Bridge: linear regressions that link ("bridge") high frequency variables to low frequency ones (e.g., monthly IP and quarterly GDP)
- MIDAS: Tightly parameterized, reduced-form regression that involves processes sampled at different frequencies; can handle large frequency mismatch by reducing the number of parameters to be estimated; it is a direct forecasting tool
- Unrestricted MIDAS (U-MIDAS): Generalization of bridge equation and MIDAS regressions; it is a linear model using OLS estimation; it cannot handle large frequency mismatch; it can be applied also to factor models

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